Blade Zero Thrust Stop System (S.O.O. 6203) – General Data

1. Description

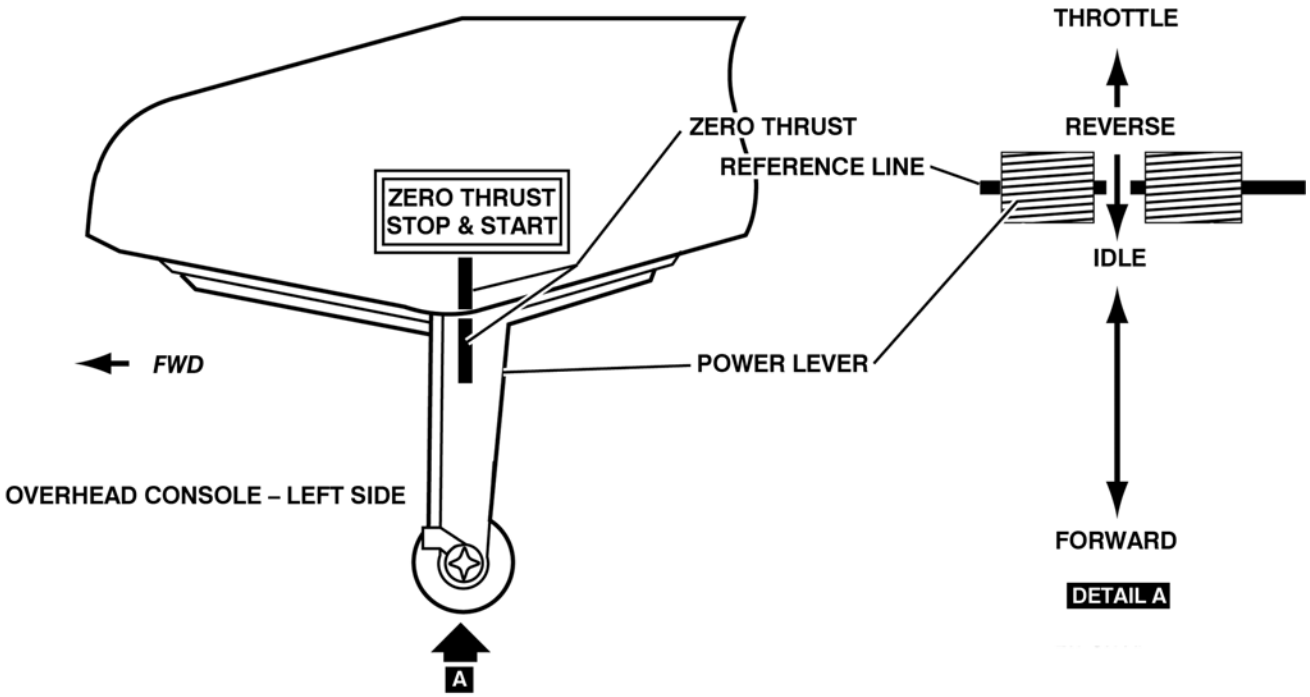
Refer to Figure 1 for details.

Each propeller is fitted with a blade zero thrust stop mechanism, which engages at a blade pitch angle of +1 degree when the engine is shutdown. This permits engine starts with zero thrust at idle power. The mechanism consists of a spring-loaded plunger and stop plate assembly installed at the root end of each blade. Reference lines on the power lever and on the overhead console, marked ZERO THRUST STOP AND START, are used to position the power lever correctly when engaging and releasing the zero thrust stops. This installation is only fitted on aircraft with floats.

2. Operation

If the zero thrust pitch stops are to be engaged on engine shutdown, the propeller lever must be set to full INCREASE, and the power lever set with the zero thrust reference lines aligned prior to stopping the engine. When the engine ceases to rotate, the oil pressure acting on the pitch change mechanism decays, due to a slow internal leakage, and the plungers engage the stop plate.

When starting an engine with blade zero thrust stops engaged, the power lever will have remained set with the zero thrust reference lines aligned and the propeller lever will be set according to OAT (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data). As the engine commences to turn, constant speed governor oil pressure rises and is directed to the propeller pitch change mechanism, which moves the blades slightly into reverse pitch, allowing the plungers to move away from the zero thrust stop plates. When the propeller speed increases to above 30% rpm, the centrifugal effect acting on the plungers moves the plungers outwards, against spring tension, to clear the stop plates. In this condition, the engine power lever can be advanced to idle and the blades move into a low forward pitch.



Location of Zero Thrust Reference Lines
Figure 1

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